Foundations of Data Science

INTRODUCTION TO DATA SCIENCE

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Definition of Data Science

Data Science is an interdisciplinary field that uses techniques from statistics, computer science, and domain knowledge to extract insights and knowledge from structured and unstructured data. It involves data collection, cleaning, analysis, visualization, and predictive modeling.

History of Data Science

Data science is a multidisciplinary field that combines statistical methods, computer science, and domain expertise to extract insights and knowledge from data. While the term “data science” is relatively new, the practices that form its foundation—such as data analysis, statistics, and computational modeling—have been developing for decades.

The evolution of data science has been shaped by the growing availability of data, advances in computing power, and the increasing need for businesses and researchers to make data-driven decisions. Over time, new tools, techniques, and roles emerged to handle larger datasets, more complex problems, and automation through machine learning and AI. This led to the formalization of data science as a distinct discipline in the early 2000s.

Timeline of Data Science

1950s–1960s: Foundations

- Statistics and mathematics were heavily used for data analysis.
- Computers began assisting in data storage and processing.
- Early research in artificial intelligence (AI) started to emerge.

1970s: Rise of Data Processing

- Businesses began using computers for data management (e.g., databases).
- The term “data processing” became common in industry.

1980s: Birth of Machine Learning

- Development of early machine learning algorithms.
- Increased focus on automating data analysis.

1990s: Data Mining Era

- The term "data mining" became popular—identifying patterns in large datasets.
- Significant advances in computing power and storage.

2001: “Data Science” Gains Recognition

- William S. Cleveland formalized data science as an independent discipline.
- The Data Science Journal was launched.
- The field began integrating computing, statistics, and domain knowledge

2010s–Present: Big Data & AI Boom

- Explosive growth of big data from social media, IoT, mobile apps, etc.
- Popular tools: Python, R, Hadoop, Spark.
- Major progress in deep learning and AI technologies.
- Data scientists became highly sought-after across industries.

Benefits of Data Science

1. **Improved Decision-Making**
   Data science helps organizations make better, data-driven decisions by uncovering trends and insights from complex data.

2. **Enhanced Customer Experience**
   By analyzing customer behavior and preferences, businesses can personalize products and services to boost satisfaction and loyalty.

3. **Increased Efficiency and Automation**
   Automating routine tasks and optimizing processes through data-driven algorithms save time and reduce costs.

4. **Competitive Advantage**
   Companies that leverage data science gain insights that help them stay ahead of competitors and identify new market opportunities.

5. **Risk Management**
   Data science can predict potential risks (e.g., fraud detection, financial risk), enabling proactive measures to minimize losses.

6.Innovation and Product Development
Insights from data enable organizations to innovate, develop new products, and improve existing offerings based on customer needs.

7.Better Marketing Strategies
Analyzing data from campaigns and customer interactions allows marketers to target the right audience effectively.

8.Healthcare Improvements
In medicine, data science supports diagnosis, personalized treatment, and prediction of disease outbreaks.

9.Resource Optimization
Organizations can optimize resource allocation, supply chain management, and operations through predictive analytics.

10.Scientific Research Advancement
Data science accelerates discoveries by enabling analysis of large-scale experimental data across various scientific fields.

Scope of Data Science

Data science covers a wide range of activities and industries, including:

- **Data Collection & Storage:** Gathering data from various sources and managing it efficiently.
- **Data Cleaning & Preparation:** Ensuring data quality by removing errors and inconsistencies.
- **Data Analysis & Visualization:** Extracting insights and presenting them in understandable formats like charts and dashboards.
- **Machine Learning & Modeling:** Building predictive models to forecast trends and automate decisions.
- **Big Data Technologies:** Handling massive datasets using tools like Hadoop and Spark.
- **Data Engineering:** Designing infrastructure and pipelines to move and process data.
- **Artificial Intelligence:** Creating intelligent systems that can learn and adapt.
- **Domain Expertise:** Applying data science methods within specific fields for targeted solutions.

Applications of Data Science

Data Science is the deep study of a large quantity of data, which involves extracting some meaning from the raw, structured, and unstructured data. Extracting meaningful data from large amounts uses algorithms processing of data and this processing can be done using statistical techniques and algorithm, scientific techniques, different technologies, etc. It uses various tools and techniques to extract meaningful data from raw data. Data Science is also known as the Future of Artificial Intelligence.

1. In Search Engines

The most useful application of Data Science is Search Engines. As we know when we want to search for something on the internet, we mostly use Search engines like Google, Yahoo, DuckDuckGo and Bing, etc. So Data Science is used to get Searches faster.

For Example, When we search for something suppose "Data Structure and algorithm courses " then at that time on Internet Explorer we get the first link of GeeksforGeeks Courses. This happens because the GeeksforGeeks website is visited most in order to get information regarding Data Structure courses and Computer related subjects. So this analysis is done using Data Science, and we get the Topmost visited Web Links.

2. In Transport

Data Science is also entered in real-time such as the Transport field like Driverless Cars. With the help of Driverless Cars, it is easy to reduce the number of Accidents.

For Example, In Driverless Cars the training data is fed into the algorithm and with the help of Data Science techniques, the Data is analyzed like what as the speed limit in highways, Busy Streets, Narrow Roads, etc. And how to handle different situations while driving etc.

3. In Finance

Data Science plays a key role in Financial Industries. Financial Industries always have an issue of fraud and risk of losses. Thus, Financial Industries needs to automate risk of loss analysis in order to carry out strategic decisions for the company. Also, Financial Industries uses Data Science Analytics tools in order to predict the future. It allows the companies to predict customer lifetime value and their stock market moves.

For Example, In Stock Market, Data Science is the main part. In the Stock Market, Data Science is used to examine past behavior with past data and their goal is to examine the future outcome. Data is analyzed in such a way that it makes it possible to predict future stock prices over a set timetable.

4. In E-Commerce

E-Commerce Websites like Amazon, Flipkart, etc. uses data Science to make a better user experience with personalized recommendations.

For Example, When we search for something on the E-commerce websites we get suggestions similar to choices according to our past data and also we get recommendations according to most buy the product, most rated, most searched, etc. This is all done with the help of Data Science.

5. In Health Care

In the Healthcare Industry data science act as a boon. Data Science is used for:

- Detecting Tumor.
- Drug discoveries.
- Medical Image Analysis.
- Virtual Medical Bots
- Genetics and Genomics.
- Predictive Modeling for Diagnosis etc.

6. Image Recognition

Currently, Data Science is also used in Image Recognition. **For Example,** When we upload our image with our friend on Facebook, Facebook gives suggestions Tagging who is in the picture. This is done with the help of machine learning and Data Science. When an Image is Recognized, the data analysis is done on one's Facebook friends and after analysis, if the faces which are present in the picture matched with someone else profile then Facebook suggests us auto-tagging.

7. Targeting Recommendation

Targeting Recommendation is the most important application of Data Science. Whatever the user searches on the Internet, he/she will see numerous posts everywhere. This can be explained properly with an example: Suppose I want a mobile phone, so I just Google search it and after that, I changed my mind to buy offline. In Real -World Data Science helps those companies who are paying for Advertisements for their mobile. So everywhere on the internet in the social media, in the websites, in the apps everywhere I will see the recommendation of that mobile phone which I searched for. So this will force me to buy online.

8. Airline Routing Planning

With the help of Data Science, Airline Sector is also growing like with the help of it, it becomes easy to predict flight delays. It also helps to decide whether to directly land into the destination or take a halt in between like a flight can have a direct route from Delhi to the U.S.A or it can halt in between after that reach at the destination.

9. Data Science in Gaming

In most of the games where a user will play with an opponent i.e. a Computer Opponent, data science concepts are used with machine learning where with the help of past data the Computer will improve its performance. There are many games like Chess, EA Sports, etc. will use Data Science concepts.

10. Medicine and Drug Development

The process of creating medicine is very difficult and time-consuming and has to be done with full disciplined because it is a matter of Someone's life. Without Data Science, it takes lots of time, resources, and finance or developing new Medicine or drug but with the help of Data Science, it becomes easy because the prediction of success rate can be easily determined based on biological data or factors. The algorithms based on data science will forecast how this will react to the human body without lab experiments.

11. In Delivery Logistics

Various Logistics companies like DHL, FedEx, etc. make use of Data Science. Data Science helps these companies to find the best route for the Shipment of their Products, the best time suited for delivery, the best mode of transport to reach the destination, etc.

The 3 V’s: Volume, Velocity, Variety

1. Volume:

- The name ‘Big Data’ itself is related to a size which is enormous.
- Volume is a huge amount of data.
- To determine the value of data, size of data plays a very crucial role. If the volume of data is very large, then it is actually considered as a ‘Big Data’. This means whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data.
- Hence while dealing with Big Data it is necessary to consider a characteristic ‘Volume’.
- *Example:* In the year 2016, the estimated global mobile traffic was 6.2 Exabytes (6.2 billion GB) per month. Also, by the year 2020 we will have almost 40000 Exabytes of data.

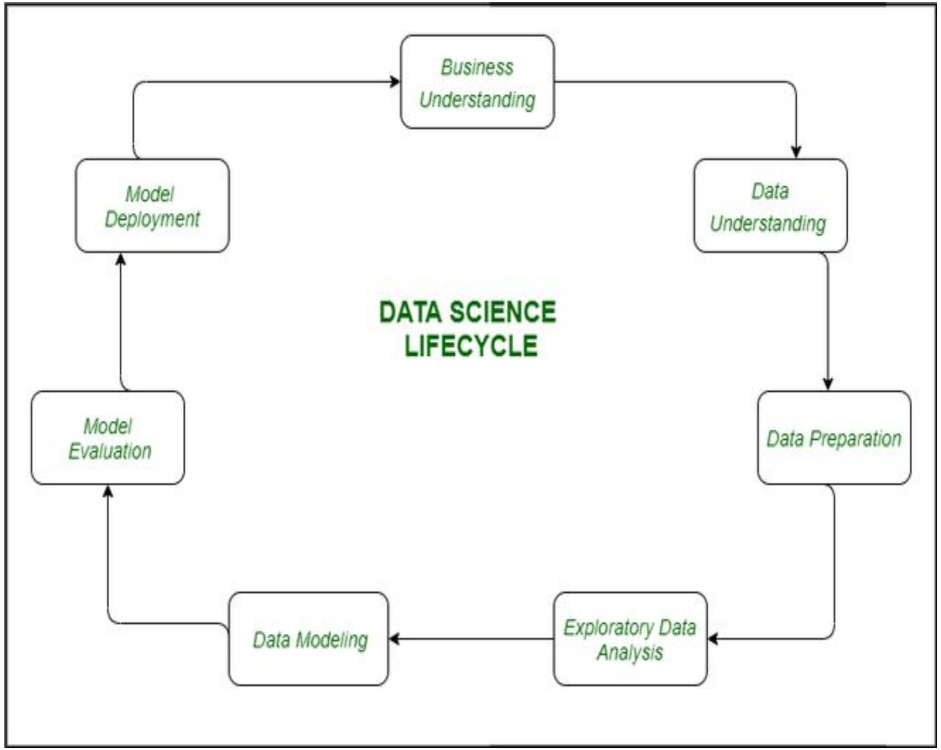
2. Velocity:

- Velocity refers to the high speed of accumulation of data.
- In Big Data velocity data flows in from sources like machines, networks, social media, mobile phones etc.
- There is a massive and continuous flow of data. This determines the potential of data that how fast the data is generated and processed to meet the demands.
- Sampling data can help in dealing with the issue like ‘velocity’.
- *Example:* There are more than 3.5 billion searches per day are made on Google. Also, Facebook users are increasing by 22%(Approx.) year by year.

3. Variety:

- It refers to nature of data that is structured, semi-structured and unstructured data.
- It also refers to heterogeneous sources.
- Variety is basically the arrival of data from new sources that are both inside and outside of an enterprise. It can be structured, semi-structured and unstructured.
    - **Structured data**: This data is basically an organized data. It generally refers to data that has defined the length and format of data.
    - **Semi- Structured data**: This data is basically a semi-organised data. It is generally a form of data that do not conform to the formal structure of data. Log files are the examples of this type of data.
    - **Unstructured data**: This data basically refers to unorganized data. It generally refers to data that doesn’t fit neatly into the traditional row and column structure of the relational database. Texts, pictures, videos etc. are the examples of unstructured data which can’t be stored in the form of rows and columns.

Data Science Lifecycle



Some steps are necessary for any of the tasks that are being done in the field of data science to derive any fruitful results from the data at hand.

- **Data Collection** - After formulating any problem statement the main task is to calculate data that can help us in our analysis and manipulation. Sometimes data is collected by performing some kind of survey and there are times when it is done by performing scrapping.
- **Data Cleaning** - Most of the real-world data is not structured and requires cleaning and conversion into structured data before it can be used for any analysis or modeling.
- **Exploratory Data Analysis** - This is the step in which we try to find the hidden patterns in the data at hand. Also, we try to analyze different factors which affect the target variable and the extent to which it does so. How the independent features are related to each other and what can be done to achieve the desired results all these answers can be extracted from this process as well. This also gives us a direction in which we should work to get started with the modeling process.
- **Model Building** - Different types of machine learning algorithms as well as techniques have been developed which can easily identify complex patterns in the data which will be a very tedious task to be done by a human.

- **Model Deployment -** After a model is developed and gives better results on the holdout or the real-world dataset then we deploy it and monitor its performance. This is the main part where we use our learning from the data to be applied in real-world applications and use cases.

Why Data Science?

- **Data science careers are in high demand.** If you choose a data science for your career path, you can rest assured that there is no shortage of opportunity for data science careers. Organizations in nearly every industry are looking for data scientists, particularly those with a robust set of skills. The demand for data scientists, coupled with the preparation our master's program offers has resulted in 100% of our students being employed within six months of graduation. Our graduates have received promotions in their current roles, lucrative job offers, and new opportunities to be part of decision-making teams.
- **There's a low supply of workers in the data science field.** Many data science positions are available, but there are very few people with the skills required to fill these roles. This means if you have storytelling, quantitative and computational data skills and can communicate effectively and act ethically, you'll have a competitive edge in securing a job in data science.
- **The data science field is versatile and broadly applicable.** Data science is a very flexible field and can be applied across all types of industries. From banking and education to marketing and nonprofits, nearly every type of organization benefits from having a data science expert on their team who can explore, report on, and monitor data to make better decisions, improve performance, and meet goals faster. So, if you're wondering what you can do with a data science degree in a specific industry, it's likely you'll find an opportunity that resonates with your career interests and values.

- **A data science career has the potential to make a lasting impact.** As a data scientist, you'll compile, analyze, and extract valuable insights from data. That data has the ability to do many powerful things: detect early-stage tumors, optimize shipping routes, improve athletic performance, and detect fraud, for starters.

Qualitative vs Quantitative Data

Qualitative Data	Quantitative Data
Qualitative data uses methods like interviews, participant observation, focus on a grouping to gain collective information.	Quantitative data uses methods as questionnaires, surveys, and structural observations to gain collective information.
Data format used in it is textual. Datasheets are contained of audio or video recordings and notes.	Data format used in it is numerical. Datasheets are obtained in the form of numerical values.
Qualitative data talks about the experience or quality and explains the questions like 'why' and 'how'.	Quantitative data talks about the quantity and explains the questions like 'how much', 'how many .
The data is analyzed by grouping it into different categories.	The data is analyzed by statistical methods.
Qualitative data are subjective and can be further open for interpretation.	Quantitative data are fixed and universal.

Data Analysis Vs. Data Analytic

Data Analytics	Data Analysis
It is described as a traditional form or generic form of analytics.	It is described as a particularized form of analytics.
It includes several stages like the collection of data and then the inspection of business data is done.	To process data, firstly raw data is defined in a meaningful manner, then data cleaning and conversion are done to get meaningful information from raw data.
It supports decision making by analyzing enterprise data.	It analyzes the data by focusing on insights into business data.
It uses various tools to process data such as Tableau, Python, Excel, etc.	It uses different tools to analyze data such as Rapid Miner, Open Refine, Node XL, KNIME, etc.
Descriptive analysis cannot be performed on this.	A Descriptive analysis can be performed on this.
One can find anonymous relations with the help of this.	One cannot find anonymous relations with the help of this.
It does not deal with inferential analysis.	It supports inferential analysis.

Data Warehousing

It is a technology that aggregates structured data from one or more sources so that it can be compared and analyzed rather than transaction processing. A **data warehouse** is designed to support the management decision-making process by providing a platform for data cleaning, data integration, and data consolidation. A data warehouse contains subject-oriented, integrated, time-variant, and non-volatile data. The Data warehouse consolidates data from many sources while ensuring data quality, consistency, and accuracy. Data warehouse improves system performance by separating analytics processing from transactional databases. Data flows into a data warehouse from the various databases. A data warehouse works by organizing data into a schema that describes the layout and type of data. Query tools analyze the data tables using schema.

Advantages of Data Warehousing

Advantages of Data Warehousing

- The data warehouse's job is to make any form of corporate data easier to understand. The majority of the user's job will consist of inputting raw data.
- The capacity to update continuously and frequently is the key benefit of this technology. As a result, data warehouses are perfect for organizations and entrepreneurs who want to stay current with their target audience and customers.
- It makes data more accessible to businesses and organizations.
- A data warehouse holds a large volume of historical data that users can use to evaluate different periods and trends in order to create predictions for the future.

Disadvantages of Data Warehousing

Disadvantages of Data Warehousing

- There is a great risk of accumulating irrelevant and useless data. Data loss and erasure are other potential issues.
- Data is gathered from various sources in a data warehouse. Cleansing and transformation of the data are required. This could be a difficult task.

Data Mining

It is the process of finding patterns and correlations within large data sets to identify relationships between data. Data mining tools allow a business organization to predict customer behavior. Data mining tools are used to build risk models and detect fraud. Data mining is used in market analysis and management, fraud detection, corporate analysis, and risk management.

Advantages and Disadvantages of Data Mining

Advantages of Data Mining

- Data mining aids in a variety of data analysis and sorting procedures. The identification and detection of any undesired fault in a system is one of the best implementations here. This method permits any dangers to be eliminated sooner.
- In comparison to other statistical data applications, data mining methods are both cost-effective and efficient.
- Companies can take advantage of this analytical tool by providing appropriate and easily accessible knowledge-based data.
- The detection and identification of undesirable faults that occur in the system are one of the most astonishing data mining techniques.

Disadvantages of Data Mining

- Data mining isn't always 100 percent accurate, and if done incorrectly, it can lead to data breaches.
- Organizations must devote a significant amount of resources to training and implementation. Furthermore, the algorithms used in the creation of data mining tools cause them to work in different ways.